

MARION COUNTY AP, AR, USA

WMO: 723447

Lat: 36.291N

Lon: 92.590W

Elev: 219

StdP: 98.72

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 03918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-7.7	-17.7	0.8	-7.6	-15.9	1.0	-5.8	9.8	9.0	8.7	4.3	1.9	310	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	36.0	24.3	34.0	23.9	32.8	23.7	25.9	32.6	25.4	32.1	24.8	31.1	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	29.5	23.3	18.6	28.6	22.8	18.0	28.0	81.6	32.5	78.7	32.3	76.1	30.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.0	5.6	-14.6	37.9	2.7	1.4	-16.5	38.9	-18.1	39.7	-19.6	40.6	-21.6	41.6		
			-14.9	26.8	2.3	0.8	-16.6	27.4	-17.9	27.9	-19.2	28.4	-20.9	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	3.1	5.1	10.2	15.3	20.3	24.8	26.1	26.2	22.3	15.3	9.4	4.7
	DBStd	9.51	6.05	6.43	5.43	4.96	3.92	2.63	2.56	2.79	3.44	4.89	5.14	5.21
	HDD10.0	727	225	163	65	13	1	0	0	0	0	8	72	179
	HDD18.3	2069	472	372	257	114	28	0	0	0	8	122	271	424
	CDD10.0	2654	11	26	70	171	320	445	499	502	370	172	53	15
	CDD18.3	954	0	1	4	23	90	195	241	244	127	28	2	1
	CDH23.3	9492	0	9	46	245	755	1964	2512	2500	1156	283	22	0
	CDH26.7	4095	0	1	6	53	228	852	1184	1222	463	86	2	0
Wind	WSAvg	2.1	2.6	2.7	2.8	2.8	1.9	1.7	1.4	1.4	1.4	1.9	2.2	2.4
Precipitation	PrecAvg	1203	81	78	109	121	124	101	90	72	119	95	119	87
	PrecMax	1697	150	155	388	237	278	194	196	147	250	276	314	142
	PrecMin	935	16	19	24	44	38	40	42	7	23	14	31	20
	PrecStd	202	41	39	79	51	52	40	43	34	69	71	77	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	24.8	27.4	31.1	32.7	36.1	37.4	38.0	34.1	31.3	26.3	21.4
		MCWB	13.6	15.6	16.4	19.5	21.9	24.7	24.6	24.1	23.1	22.1	18.0	15.0
	2%	DB	18.7	21.3	24.1	27.8	31.0	33.9	35.1	36.3	32.8	29.0	22.9	18.9
		MCWB	13.6	14.4	15.6	17.4	22.1	23.9	24.7	24.0	23.0	21.2	15.3	14.1
	5%	DB	16.2	18.7	22.3	26.3	29.0	32.6	33.7	34.1	31.8	27.0	21.2	16.2
		MCWB	10.7	12.9	14.4	16.7	21.6	23.7	24.4	23.7	22.7	19.2	15.0	11.1
	10%	DB	12.8	16.3	20.0	23.9	27.5	31.3	32.5	32.7	29.4	23.9	18.7	13.0
		MCWB	8.1	11.1	13.3	15.9	20.3	23.2	24.1	23.7	22.0	17.8	13.6	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	17.7	18.8	20.7	24.4	26.2	26.8	26.9	25.3	23.7	19.8	17.0
		MCDB	19.5	23.0	25.0	27.2	30.4	33.3	33.0	33.7	31.7	29.3	24.6	20.0
	2%	WB	14.8	15.8	16.9	19.4	23.3	25.3	26.0	25.9	24.4	22.3	17.6	14.9
		MCDB	17.9	20.1	23.0	25.2	28.9	32.2	32.6	32.5	30.3	27.6	21.4	18.4
	5%	WB	11.4	14.1	15.7	18.4	22.4	24.7	25.4	25.2	23.7	20.7	15.7	12.1
		MCDB	15.0	17.9	20.4	24.0	27.6	31.0	32.0	32.1	29.1	24.5	19.6	14.6
	10%	WB	8.4	11.5	14.3	17.2	21.5	23.9	24.9	24.4	22.8	19.1	13.9	10.2
		MCDB	12.3	15.5	18.7	22.7	26.1	29.7	31.0	30.7	27.8	22.9	17.8	13.3
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.4	12.2	12.5	11.4	11.3	11.3	11.5	11.9	12.7	12.7	10.6
		MCDBR	15.6	15.6	15.3	15.1	13.5	13.1	12.6	13.9	14.2	15.0	16.1	14.4
		MCWBR	10.3	10.3	8.9	7.5	6.3	5.2	4.6	4.7	5.7	7.6	10.1	10.2
	5% WB	MCDBR	12.8	13.1	13.0	12.0	11.6	11.6	11.3	11.9	11.7	12.7	12.8	12.4
		MCWBR	10.0	9.9	8.6	7.1	5.9	5.1	4.7	4.6	5.3	7.2	9.4	10.0
Clear-Sky Solar Irradiance	taub	0.309	0.315	0.350	0.386	0.421	0.438	0.468	0.466	0.398	0.353	0.323	0.301	
	taud	2.494	2.474	2.413	2.316	2.277	2.244	2.217	2.195	2.371	2.479	2.514	2.533	
	Ebn at Noon	889	927	917	897	865	847	818	812	856	870	865	877	
	Edh at Noon	82	94	109	127	134	138	141	142	113	93	80	74	
All-Sky Solar Radiation	RadAvg	2.43	2.99	4.06	5.14	5.64	6.45	6.20	5.68	4.86	3.67	2.62	2.07	
	RadStd	0.20	0.37	0.26	0.33	0.49	0.45	0.36	0.37	0.39	0.45	0.29	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page